

Prompts for AI Supply Chain Analytics – Data Processing & KPI Framework

Step 1: Create Date Table

- Create a date table from **March 1, 2025 → May 31, 2025**
 - Ensure continuous daily records for time-series analysis
 - Include columns: date, day, month, year, week, quarter
 - This table will act as a reference for joins and reporting
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Step 2: Create Exchange Rate Table

- Create exchange_rate table from **March 1, 2025 → May 17, 2025**
 - Use API format:
 - `response = requests.get(f'https://openexchangerates.org/api/historical/YYYY-MM-DD.json?app_id=<your_app_id>&symbols=INR')`
 - Replace YYYY-MM-DD dynamically for each date
 - Store fields: date, USD_INR_Rate
 - Enables conversion of USD transactions into INR
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Step 3: Data Cleaning & Fact Table (fact_summary)

Load Data:

- fact_order_line table
- dim_products sheet (skip 1st row, 2nd row as header)
- dim_customers sheet (skip 1st row, 2nd row as header)
- exchange_rate sheet (skip 1st row, 2nd row as header)

Cleaning:

- Convert product_id, customer_id → numeric
- Strip whitespace from IDs

- Remove rows with NULL IDs
- Convert IDs → integers
- Convert date columns → datetime

Merging:

- Orders + Products using product_id
- Result + Customers using customer_id
- Result + Exchange Rates using order_placement_date

Calculations:

- For USD: $\text{price_USD} * \text{USD_INR_Rate} * \text{order_qty}$
- For INR: $\text{price_INR} * \text{order_qty}$
- Create unified column: total_amount (INR)

Final Output Cleanup:

- Drop: price_USD, price_INR, currency, Date, USD_INR_Rate
 - Keep: order details, IDs, dates, quantities, delivery info, total_amount
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Step 4: Business KPIs

- **Total Order Lines:** total number of rows
 - **Line Fill Rate:** fulfilled lines / total lines
 - **Volume Fill Rate:** delivered quantity / ordered quantity
 - **Total Orders:** count of unique orders
 - **On-Time Delivery % (OT):** on-time deliveries / total deliveries
 - **In-Full Delivery % (IF):** complete deliveries / total deliveries
 - **OTIF %:** orders delivered both on-time and in-full
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Step 5: Top Customers

- Identify top 5 customers based on total order value
- Aggregate order value at customer level
- Include columns:

- Customer Name
 - Customer ID
 - City
 - OT %, IF %, OTIF %
 - Helps identify high-value and high-performance customers
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Step 6: Top Customers (India)

- Filter dataset where country = India
- Rank customers by total order value
- Select top 5 customers
- Include:
 - Customer Name
 - Customer ID
 - City
 - OT %, IF %, OTIF %
- Provides region-specific customer insights